Nihal Ahmed

PhD Candidate in Agricultural Economics and Management

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Research Profile

PhD researcher focusing on agricultural systems, ecological sustainability, and planetary boundaries. My research quantifies trade-embodied ecological pressures, with particular emphasis on ecosystem loss, land degradation, climate stress, and ecological equity. I combine multi-regional input–output modelling, spatial analysis, machine learning, and scenario modelling to investigate the links between consumption, trade, and ecological limits.

Education

PhD in Agricultural Economics and Management *Sep 2022 – Dec 2026*

Northwest A&F University, China

Supervisor: Prof. Huai Jianjun

Master's in Applied Economics *2019 – 2022*

Kohat University of Science and Technology, Pakistan

Research Experience

PhD Researcher *Sep 2022 – Present*

Northwest A&F University, China

Trade-Embodied Ecosystem Accounting

- Developed consumption-based accounting frameworks to quantify belowground ecosystem multifunctionality (BEMF) embodied in agricultural trade.
- Integrated spatial soil-function data with agricultural production and trade data to trace ecosystem pressures from production regions to final consumption across 192 regions and 123 sectors.
- Quantified ecological pressures associated with China's agricultural imports and evaluated their distribution across exporting regions under climate change.

Deep Learning for Land-Use Scenario Synthesis

- Developed generative deep-learning models, including Variational Autoencoder (VAE), Conditional GAN (CGAN), and Recurrent Conditional GAN (RCGAN), to synthesize global land-use trajectories from CMIP6/LUH2 scenarios.
- Generated 30,000 physically plausible land-use pathways and developed a computationally efficient approach for exploring large scenario spaces.
- Combined Random Forest and SHAP analysis to identify major drivers of land degradation risk under SDG 15.3.1.

Provincial Ecological Efficiency and Spillover Networks

- Evaluated the efficiency of China's 31 provinces in transforming agricultural innovation into land ecological security using Slacks-Based Measure (SBM) efficiency models with undesirable outputs.
- Constructed interprovincial spillover networks using a modified gravity model and applied CONCOR block modelling and MRQAP to investigate structural and functional patterns of ecological spillovers.

- Identified factors associated with interprovincial differences and spillover relationships in agricultural ecological security.

Selected Publications

Belowground ecosystem multifunctionality embodied in China's agricultural imports: thermal geography, equity trajectory, and a multi-indicator accounting framework.
N. Ahmed, N.A. Khan, J. Ober, P. Sakiewicz, J. Huai.
Ecological Indicators, 2026. doi:10.1016/j.ecolind.2026.115653 **IF:** 8.7 (JCR 2025)

Agricultural trade externalizes belowground ecosystem loss to climate-stressed exporting nations.
N. Ahmed, N.A. Khan, J. Ober, J. Huai.
International Journal of Sustainable Development & World Ecology, 2026. doi:10.1080/13504509.2026.2726387 **IF:** 6.8 (JCR 2025)

Deep learning approaches for synthesizing Land Ecological Security pathways in global land-use scenarios.
N. Ahmed, N.A. Khan, J. Ober, M.F. Shahzad.
Resources, Environment and Sustainability, 25, 100345, 2026. doi:10.1016/j.resenv.2026.100345 **IF:** 9.2 (JCR 2025)

Efficiency divergence in agricultural innovation-to-ecological security transformation: Network analysis of Chinese provincial spillovers.
N. Ahmed, J. Ober, J. Huai.
Environmental and Sustainability Indicators, 2026. doi:10.1016/j.indic.2026.101224 **IF:** 5.5 (JCR 2025)

Quantifying China's ecological assets: A carbon-based assessment of climate change and land use change effects using generalized additive models.
N. Ahmed, N.A. Khan, J. Ober, J. Huai.
Sustainable Futures, 10, 101350, 2025. doi:10.1016/j.sftr.2025.101350 **IF:** 6.7 (JCR 2025)

Synergistic evaluation of energy security and environmental sustainability in BRICS geo-political entities: An integrated index framework.
N. Ahmed, F. Ore Areche, J.M. More López, V.G. Sánchez Araujo, J.A. Legua Cárdenas, J. Ober.
Equilibrium. Quarterly Journal of Economics and Economic Policy, 19(3), 793–839, 2024. doi:10.24136/eq.3088 **IF:** 6.7 (JCR 2025)

Publication record: 19 publications published or in press; 16 first-author publications.
Full publication list: Google Scholar | nihalahmed.uk

Research Methods & Technical Skills

Environmental & Trade Modelling

- Multi-Regional Input–Output (MRIO) analysis
- Consumption-based ecological accounting
- Structural Decomposition Analysis (SDA)
- Ecosystem multifunctionality assessment
- Agricultural and ecological footprint analysis

Machine Learning & Computational Modelling

- Variational Autoencoder (VAE)
- Conditional GAN (CGAN)
- Recurrent Conditional GAN (RCGAN)
- Random Forest
- XGBoost
- SHAP / Explainable AI
- Generalized Additive Models (GAM)

- Monte Carlo simulation

Efficiency, Econometrics & Network Analysis

- SBM efficiency analysis with undesirable outputs
- Simar–Wilson bootstrap
- Panel econometrics
- Two-Way Fixed Effects
- Spatial Durbin Models
- Panel Granger Causality
- Social Network Analysis
- Modified Gravity Models
- MRQAP

Software

R | **Python** | **Stata** | **MATLAB** | **Adobe Illustrator**

Environmental & Earth-System Data

CMIP6 | **LUH2** | **FABIO** | **SPAM** | **SoilGrids** | **WorldClim** | **ESA CCI** | **MODIS**

Research Funding

National Natural Science Foundation of China (NSFC)
Grant No. 42075172
Principal Investigator: Prof. Huai Jianjun
Role: Graduate Research Team Member

Teaching Experience

Lecturer *2021 – 2022*
Government College of Technology, Pakistan

Academic Service

Peer Reviewer

- Completed 115+ manuscript reviews for 22 international journals.
- Reviewer for journals including *Ecological Indicators*, *Agricultural Systems*, and *Journal of Environmental Management*.

References

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Silesian University of Technology, Gliwice, Poland
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Prof. Huai Jianjun
Professor; Doctoral Supervisor
School of Economics and Management
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